Query Optimization: Transformation of Relational Expressions DBMS

> **Definition: Query Optimization** is the component of a Database Management System (DBMS) that attempts to determine the most efficient execution plan for evaluating a given query by transforming relational algebra expressions into equivalent, lower-cost evaluation trees.

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1. Detailed Technical Explanation

Query Processing Steps:

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### Key Relational Algebra Equivalence Rules:

1. **Commutativity of Selection:**

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### Heuristic Optimization Algorithm:

1. Break down complex query conditions into simple selections.

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## 2. Core Concepts & Memory Keywords

- **Equivalence Rules:** Algebraic identities ensuring two query trees return identical tuple results.

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3. Must-Write Points for Exams

- Query optimization chooses the physical execution path with minimal disk I/O cost.

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4. Quick Recall Flow

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